In [ ]:
```
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```
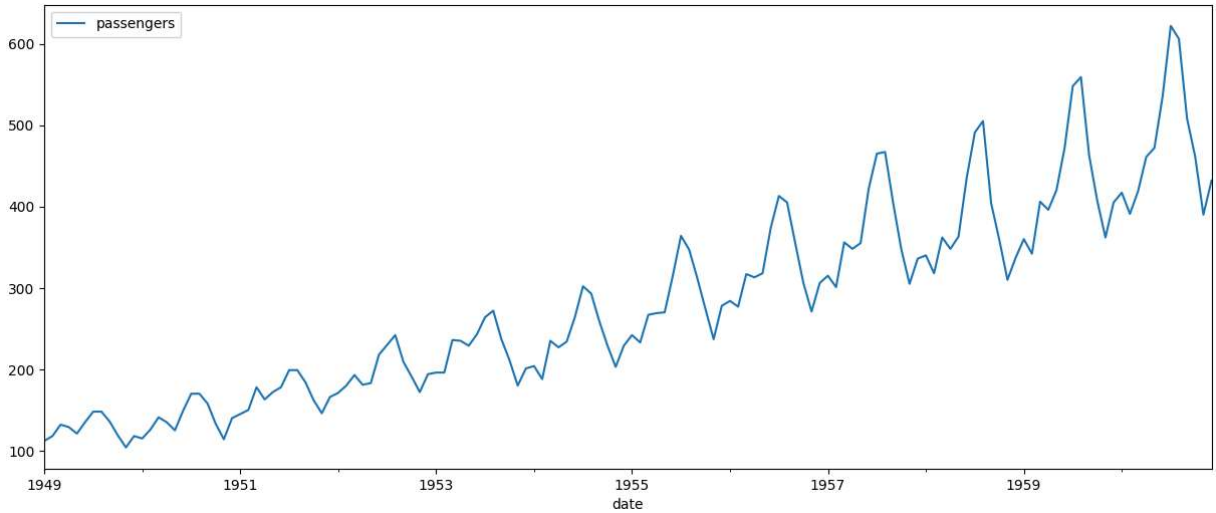
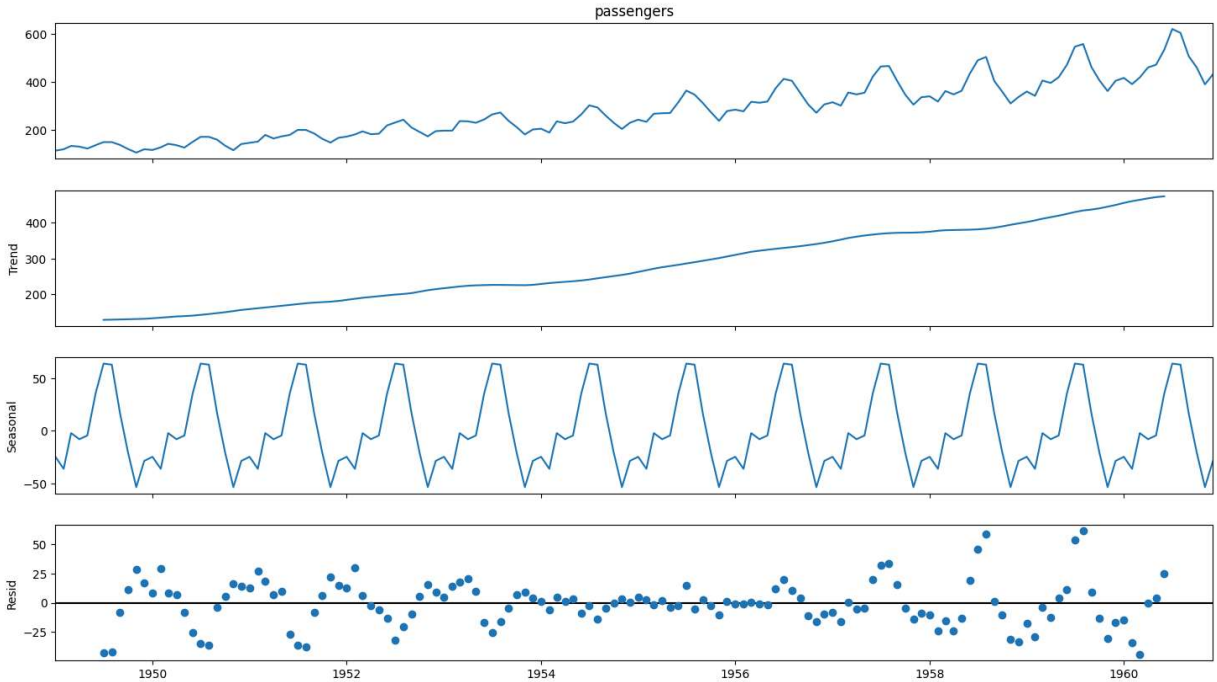

In [1]:
```
from datetime import datetime
import pandas as pd
import matplotlib.pyplot as plt
from statsmodels.tsa.seasonal import seasonal_decompose
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
from statsmodels.tsa.statespace.sarimax import SARIMAX
import numpy as np
```

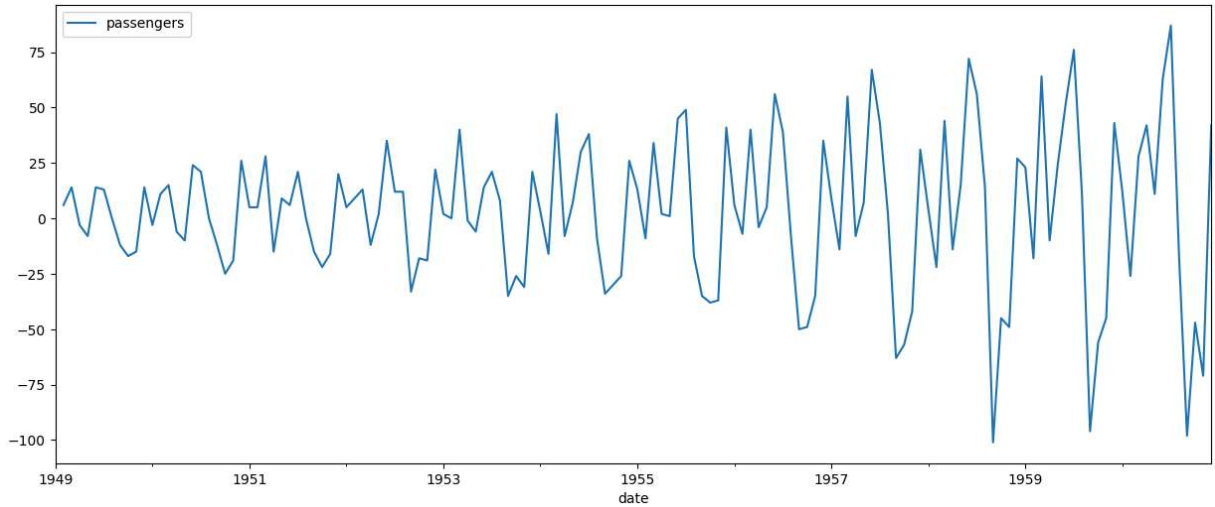
In [2]:
```
data = pd.read_csv('air_passengers.csv', sep=',')
df = pd.DataFrame(data, columns=['date', 'passengers'])

# Set the Date as Index
df['date'] = pd.to_datetime(df['date'])
df.index = df['date']
del df['date']

df.plot(figsize=(15, 6))
plt.show()
```



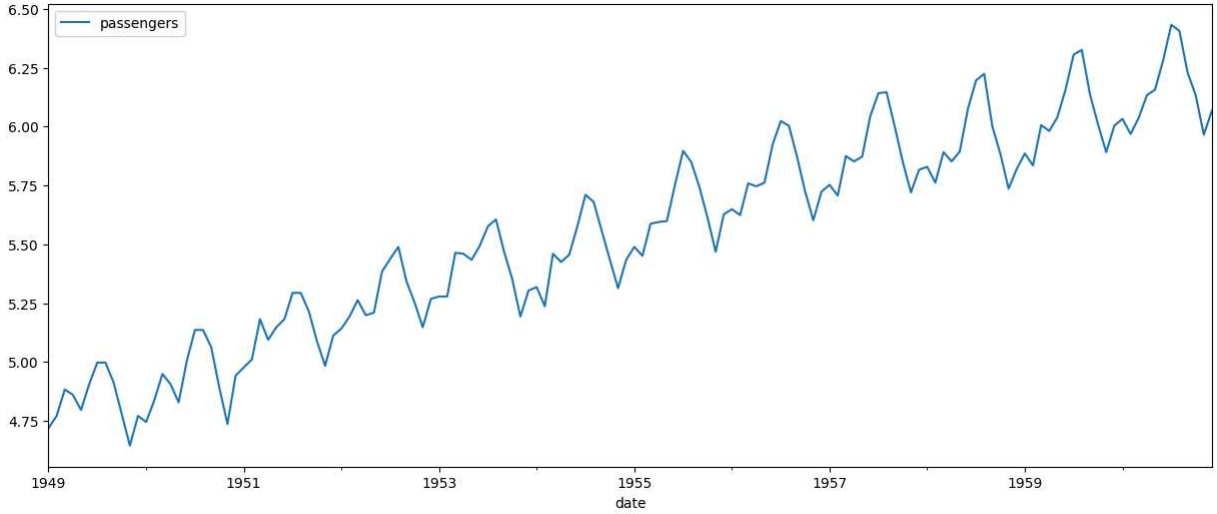
In [3]:
```
decompose = seasonal_decompose(df['passengers'], model='additive', period=12)

decompose_graph = decompose.plot()
decompose_graph.set_size_inches(16, 9)

plt.show()
```



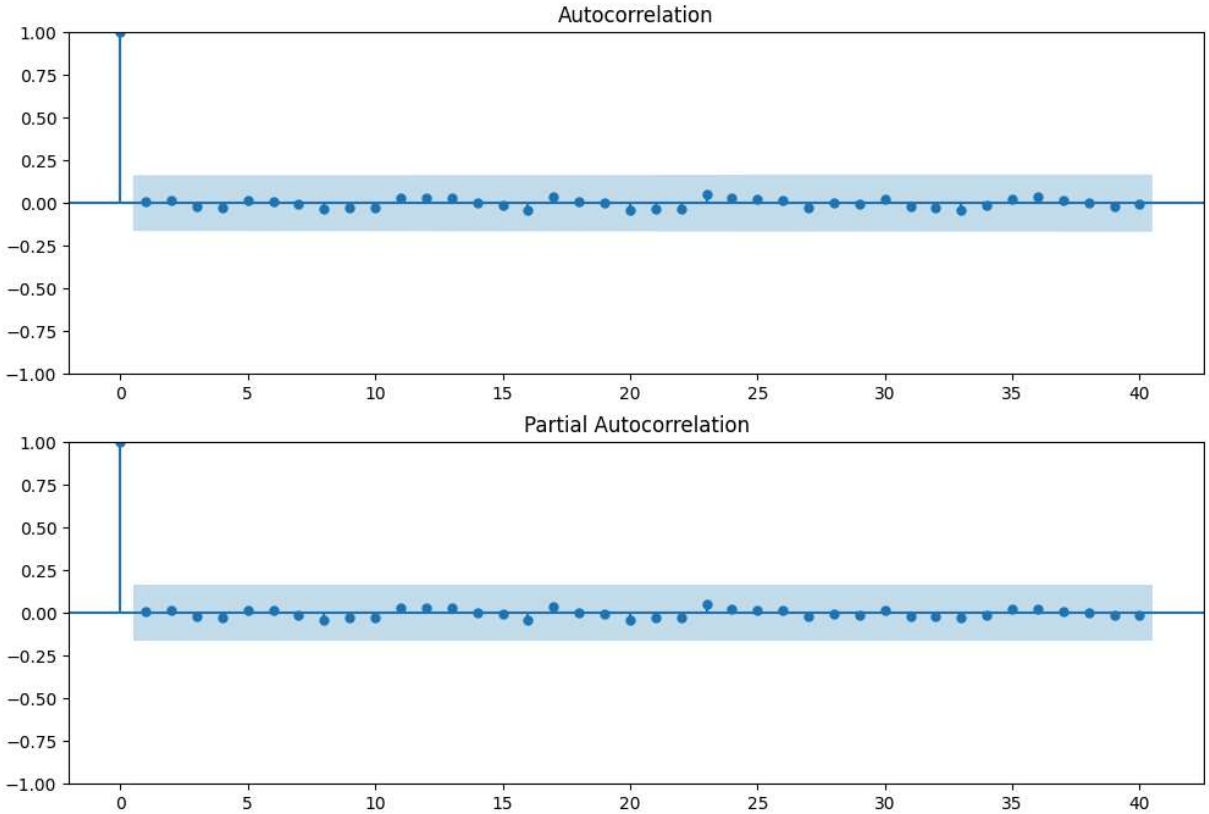
In [4]:
```
differenced_df = df.diff()

differenced_df.plot(figsize=(15, 6))
plt.show()
```



In [5]:
```
df_log = np.log(df)

df_log.plot(figsize=(15, 6))
plt.show()
```



In [8]:
```
fig = plt.figure(figsize=(12, 8))

ax1 = fig.add_subplot(211)
plot_acf(model.resid, lags=40, ax=ax1)

ax2 = fig.add_subplot(212)
plot_pacf(model.resid, lags=40, ax=ax2)

plt.show()
```



In [9]:
```
arima_model = SARIMAX(df_log, trend='n', order=(1, 1, 1), seasonal_order=(1, 1, 1,
model = arima_model.fit()

print(model.summary())
```

```
C:\Users\jdele\anaconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473: Va
lueWarning: No frequency information was provided, so inferred frequency MS will be
used.
  self._init_dates(dates, freq)
C:\Users\jdele\anaconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:473: Va
lueWarning: No frequency information was provided, so inferred frequency MS will be
used.
  self._init_dates(dates, freq)
```

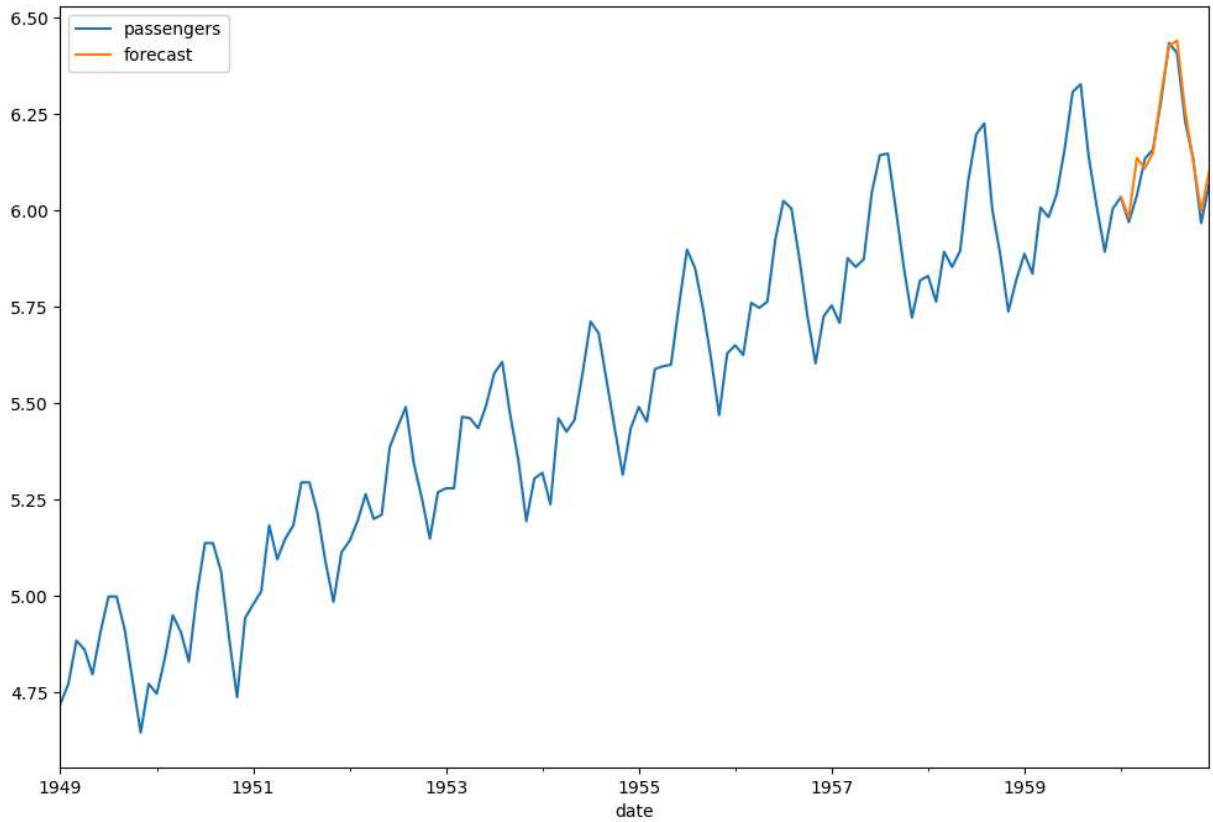
```
                                     SARIMAX Results
==================================================================================
======
Dep. Variable:                        passengers   No. Observations:
144
Model:             SARIMAX(1, 1, 1)x(1, 1, 1, 12)   Log Likelihood                 2
45.151
Date:                            Mon, 04 May 2026   AIC                           -4
80.303
Time:                                    19:35:27   BIC                           -4
65.927
Sample:                                01-01-1949   HQIC                          -4
74.461
                                     - 12-01-1960
Covariance Type:                              opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1          0.1615      0.213      0.757      0.449      -0.257       0.580
ma.L1         -0.5565      0.185     -3.002      0.003      -0.920      -0.193
ar.S.L12      -0.1003      0.196     -0.510      0.610      -0.485       0.285
ma.S.L12      -0.4962      0.210     -2.366      0.018      -0.907      -0.085
sigma2         0.0013      0.000      8.488      0.000       0.001       0.002
===================================================================================
Ljung-Box (L1) (Q):                   0.01   Jarque-Bera (JB):                 3.38
Prob(Q):                              0.91   Prob(JB):                         0.18
Heteroskedasticity (H):               0.61   Skew:                            -0.01
Prob(H) (two-sided):                  0.11   Kurtosis:                         3.79
===================================================================================

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-ste
p).
```

In [10]:

```python
#Predicciones
predictions = df_log.copy()

predictions['forecast'] = model.predict(start=132, end=143, dynamic=True)

predictions[['passengers', 'forecast']].plot(figsize=(12, 8))
```

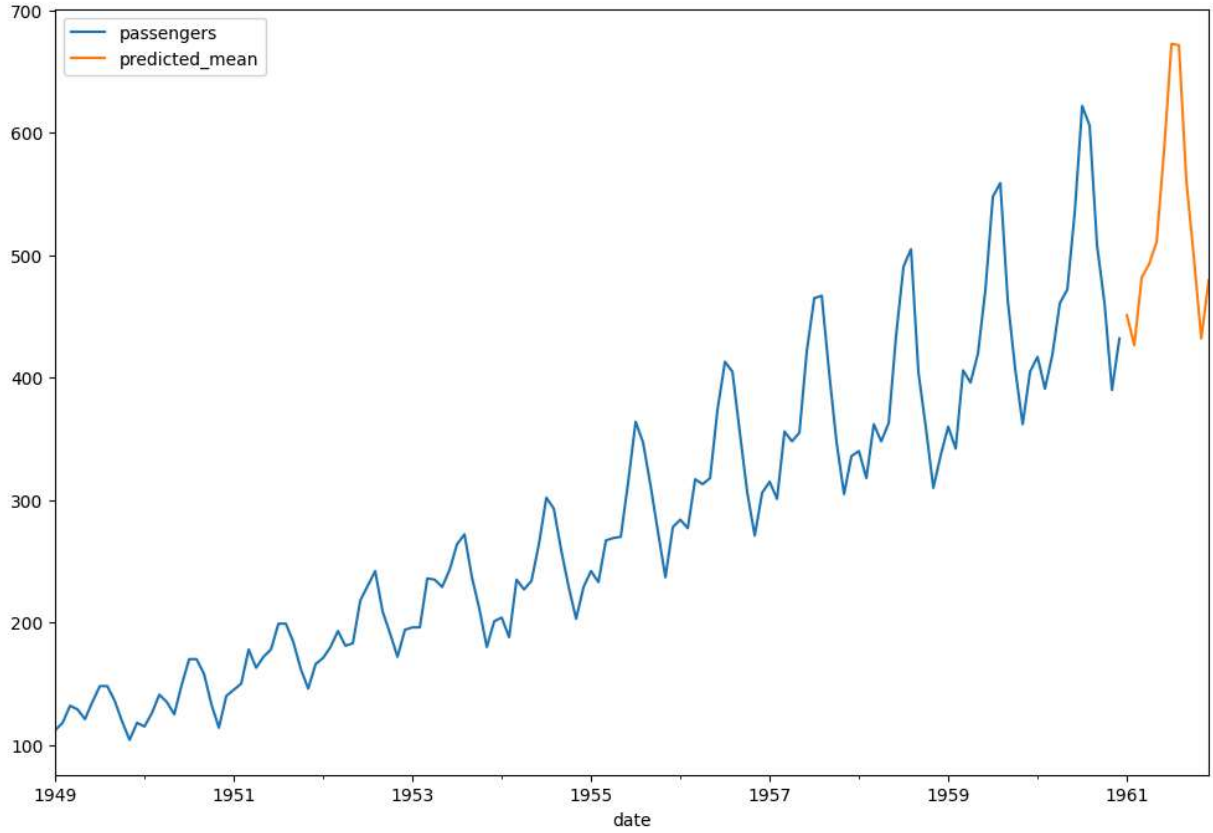
Out[10]:

```
<Axes: xlabel='date'>
```



In [11]:
```
forecast = model.forecast(steps=12)

np.exp(df_log).plot(figsize=(12, 8), legend=True)
np.exp(forecast).plot(figsize=(12, 8), legend=True)
```

Out[11]: <Axes: xlabel='date'>


passengers
predicted_mean
700
600
500
400
300
200
100
1949
1951
1953
1955
1957
1959
1961
date

In [ ]: